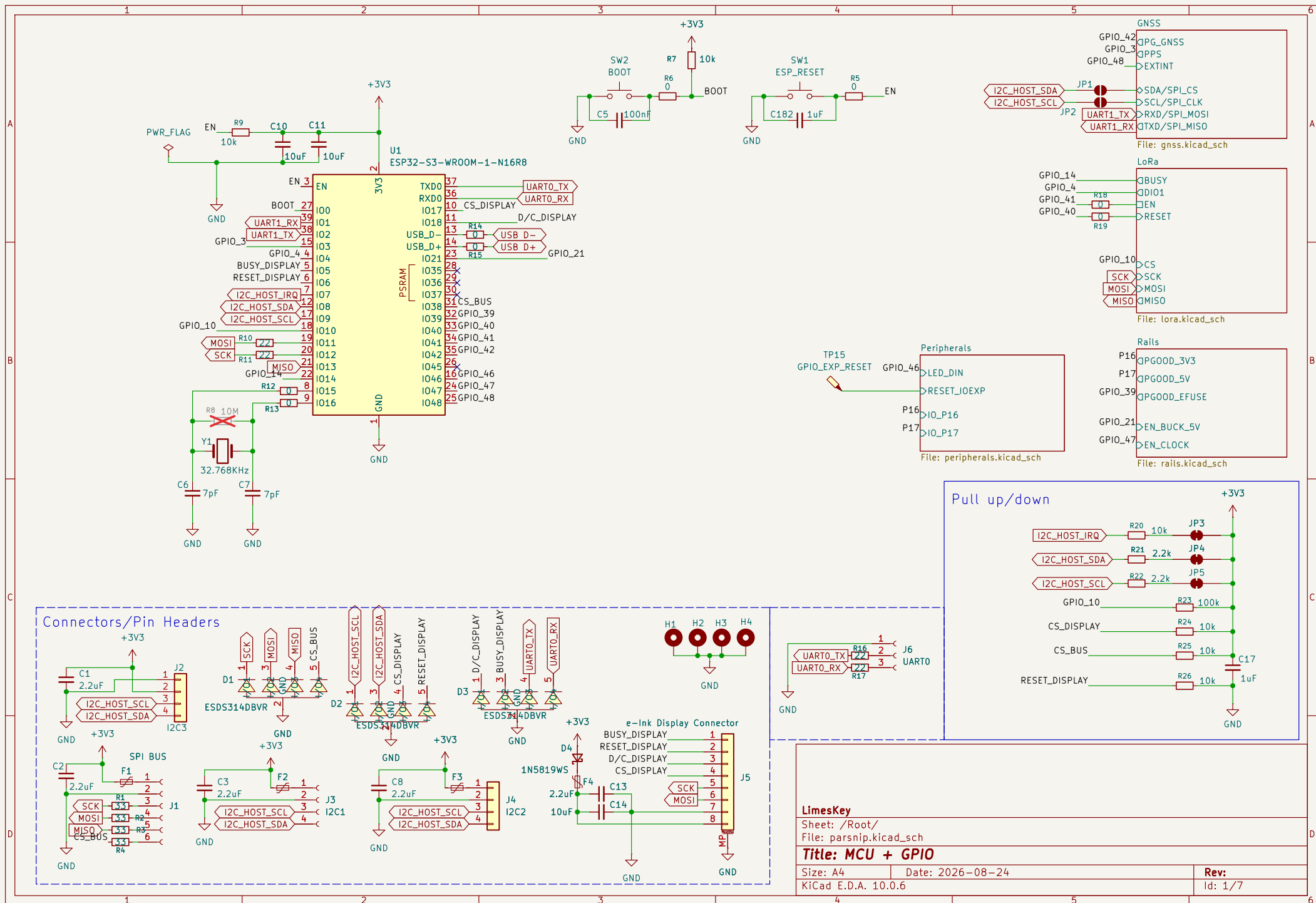
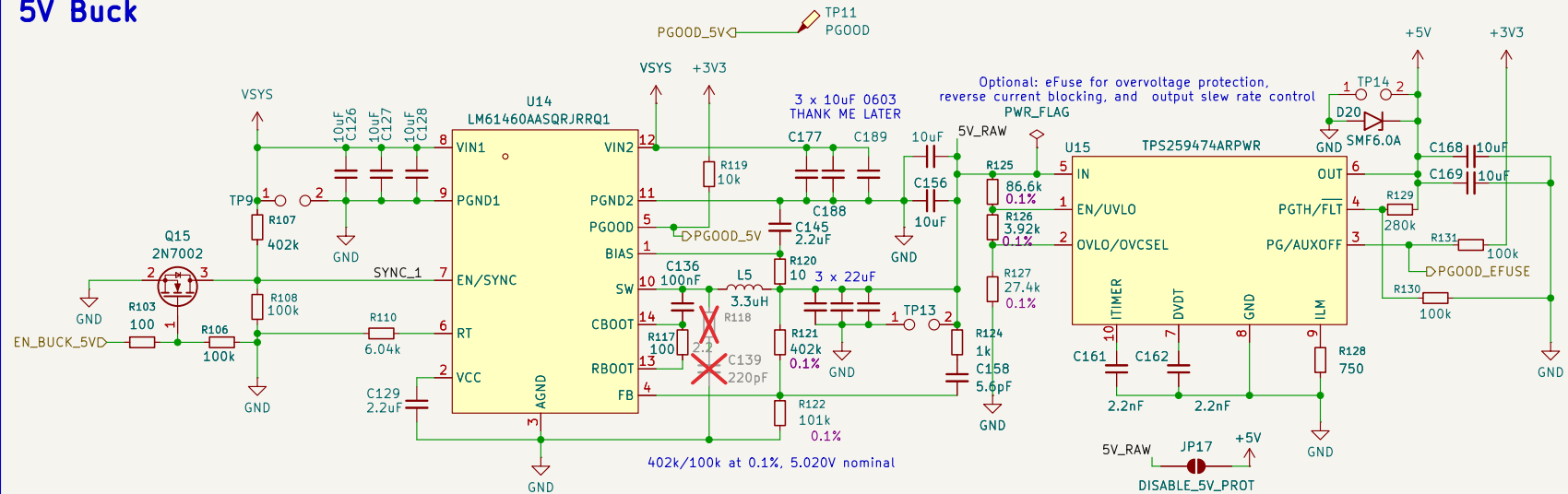
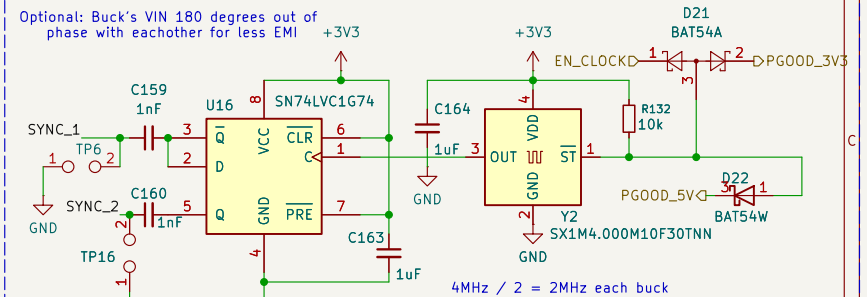
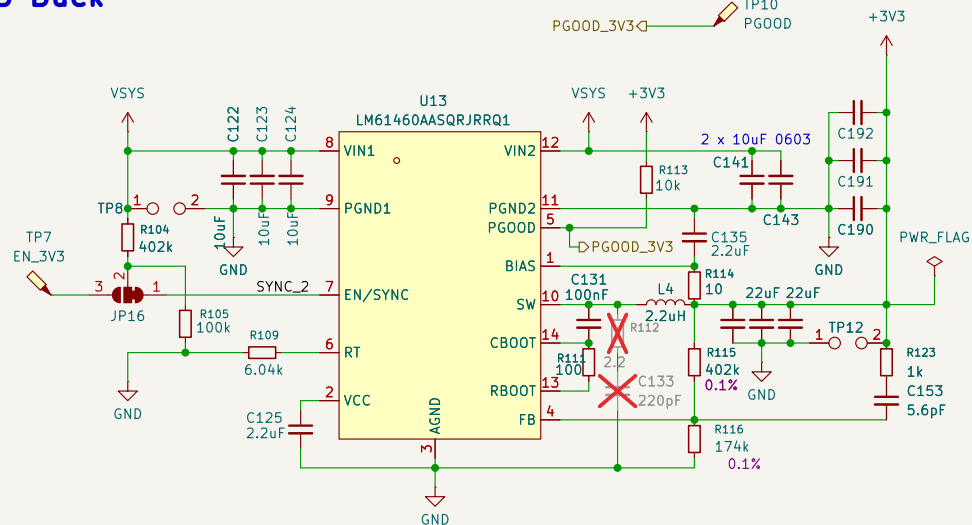




5V Buck



3V3 Buck



LimesKey

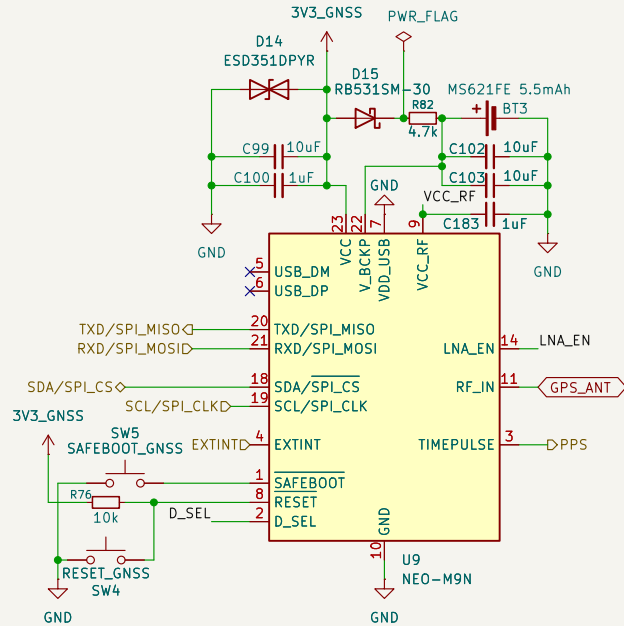
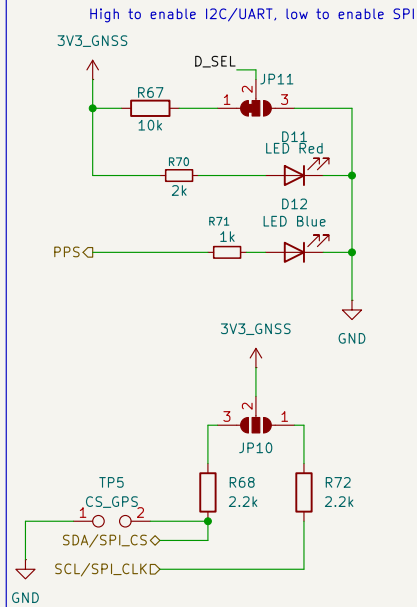
Sheet: /Root/Rails/
File: rails.kicad_sch

Title: Power Regulators

Size: A4 Date: 2026-08-24
KiCad E.D.A. 10.0.6

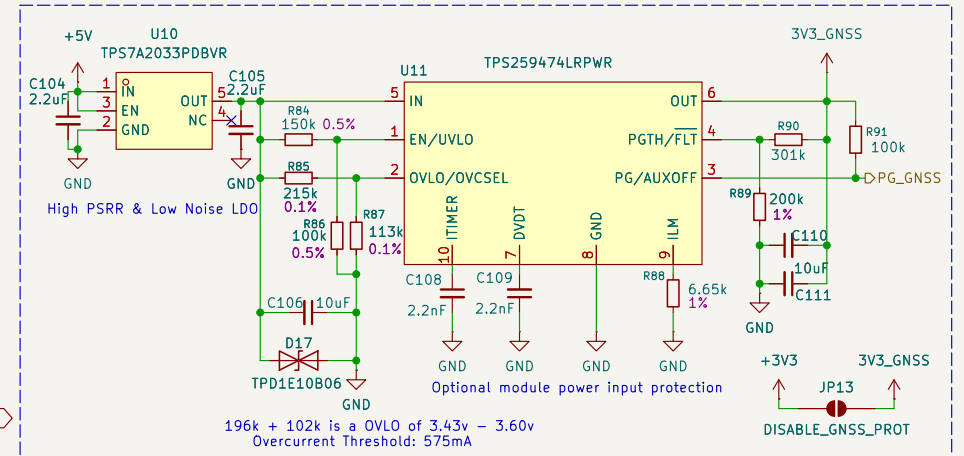
Rev:
Id: 4/7

LEDs and Jumpers

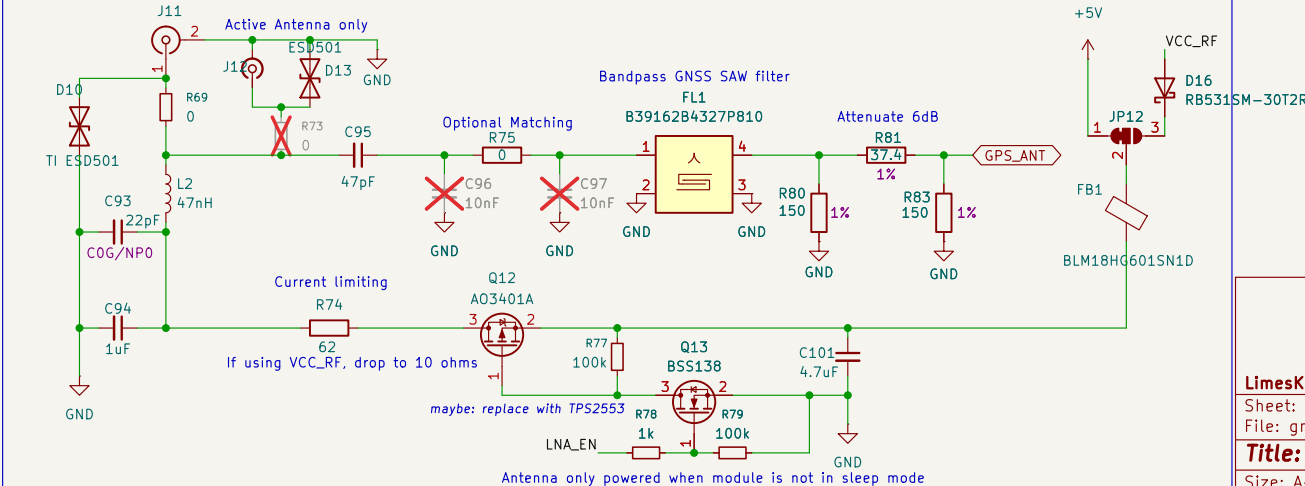


todo: switch module to use UART and not SPI
Theoretically compatible with NEO-M9N/M8N/M7N/M6N

Sparkfun Schematic: https://cdn.sparkfun.com/assets/1/0/a/8/a/SparkFun_GPS_NEO_M9N_SMA_UPDATE.pdf
U-Blox Datasheet: https://content.u-blox.com/sites/default/files/NEO-M9N_Integrationmanual_UBX-19014286.pdf



SMA + IPEX Connector



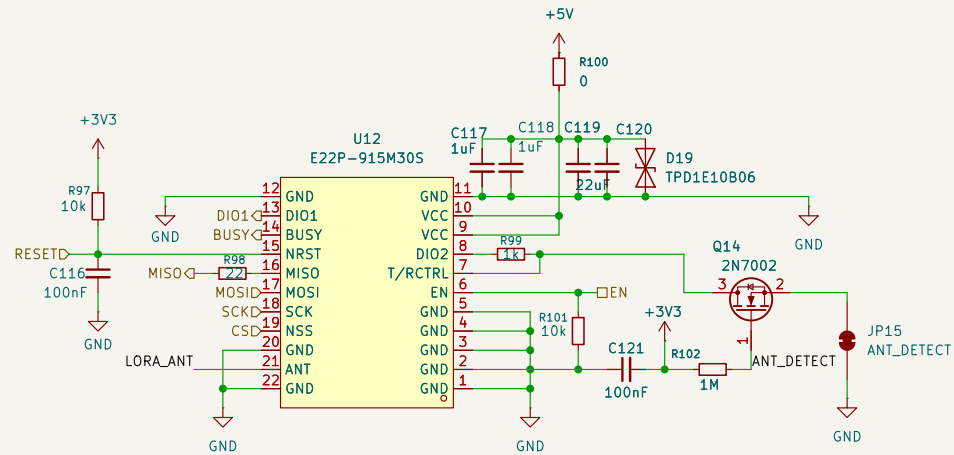
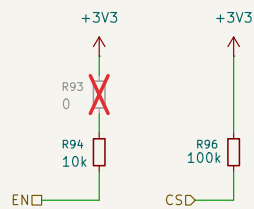
LimesKey

Sheet: /Root/GNSS/
File: gnss.kicad_sch

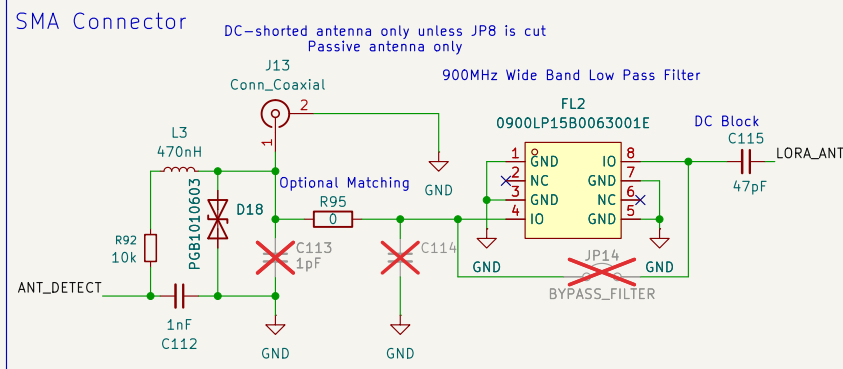
Title: GNSS Module / Active RF Antenna

Size: A4 Date: 2026-05-14
KiCad E.D.A. 10.0.6

Rev:
Id: 5/7



Ensure you set DIO2 as RF Switch in program
https://jgromes.github.io/RadioLib/class_s_x126x.html#ae46e08d579f4acbad029b4cd4f4ffaf



LimesKey

Sheet: /Root/LoRa/
 File: lora.kicad_sch

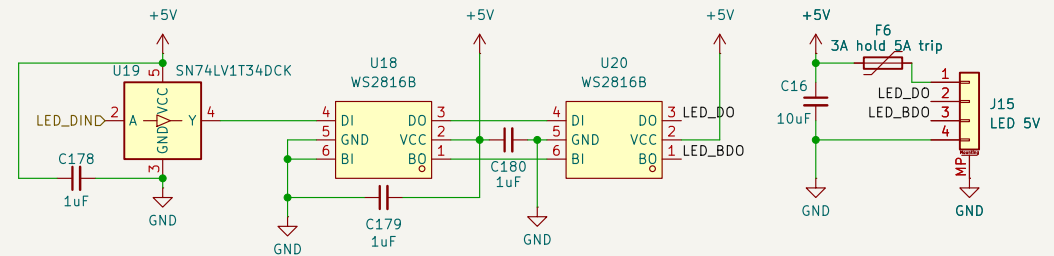
Title: LoRa Module / LoRa Antenna

Size: A4 Date: 2026-08-24
 KiCad E.D.A. 10.0.6

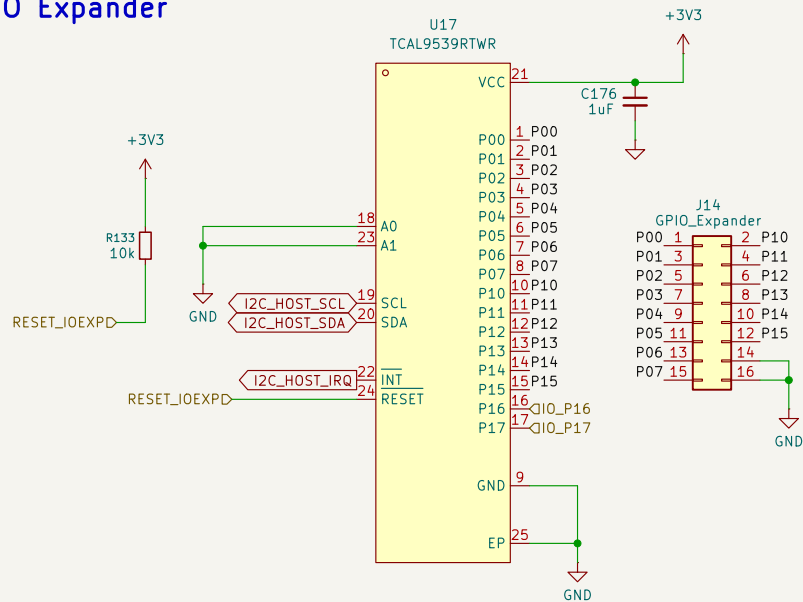
Rev:
 Id: 6/7

Individually Addressable RGB LEDs

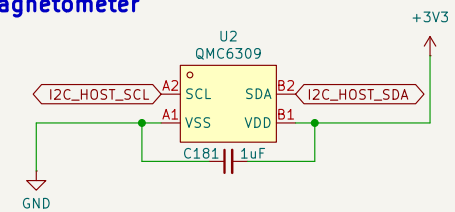
For use with WS2813/WS2816



GPIO Expander



Magnetometer



LimesKey

Sheet: /Root/Peripherals/
File: peripherals.kicad_sch

Title: GPIO Expander / LEDs / Magnetometer

Size: A4 Date: 2026-08-30

KiCad E.D.A. 10.0.6

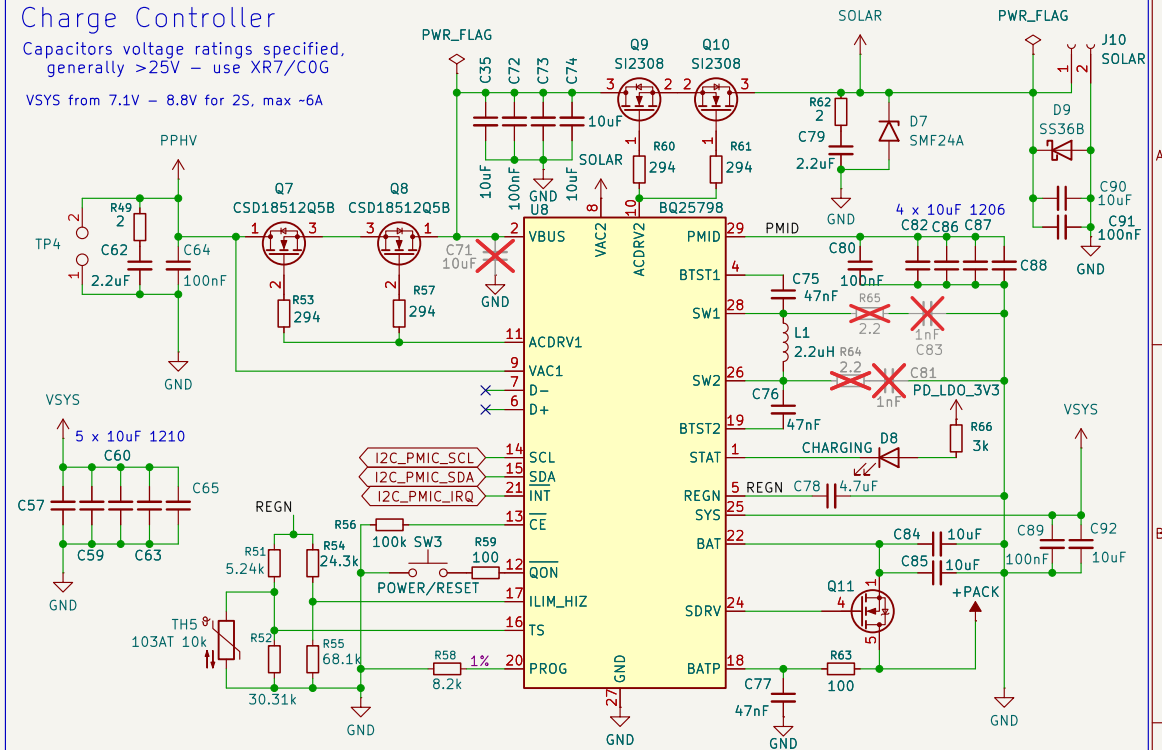
Rev:

Id: 7/7

Charge Controller

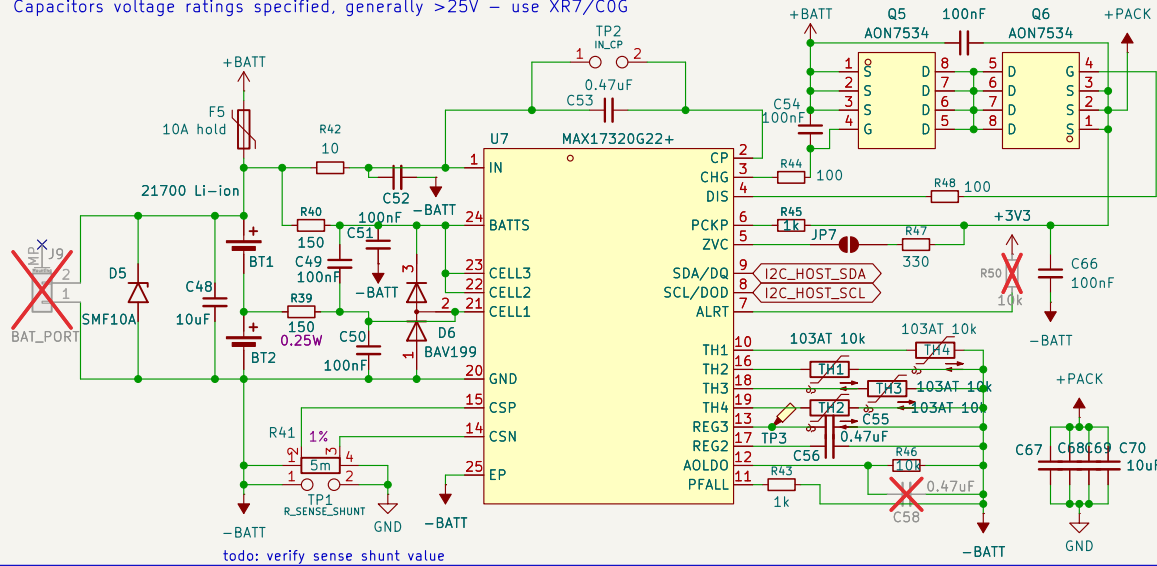
Capacitors voltage ratings specified, generally >25V – use XR7/COG

VSYS from 7.1V – 8.8V for 2S, max -6A



Battery Management System (BMS)

Capacitors voltage ratings specified, generally >25V – use XR7/COG



LimesKey

Sheet: /Charger/
File: battery.kicad_sch

Title: Battery Charger / Battery Management System (BMS)

Size: A4 Date: 2026-08-24

KiCad E.D.A. 10.0.6

Rev:
Id: 3/7